# “DEVELOPMENT AND VALIDATION STABILITY INDICATING RP-HPLC AND UV SPECTROPHOTOMETRIC METHOD FOR DETERMINATION OF LENALIDOMIDE IN BULK AND TABLET”

**A Thesis submitted**
**In Partial fulfillment of the requirements**
**For the Degree of**

## MASTER OF PHARMACY
**In**
**PHARMACEUTICAL CHEMISTRY**
**By**


## ASHUKA SINGH
**Enrollment No.** 201020050072230

**Under the Supervision of**

**DR. ROHIT TRIPATHI**
**(PROFESSOR)**

## SHAMBHUNATH INSTITUTE OF PHARMACY, JHALWA, PRAYAGRAJ


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## List of Symbols/Abbreviations

| Abbreviations/Symbols | Full Form |
|---|---|
| **%** | Percentage |
| **%RSD** | Percentage relative standard deviation |
| **±** | Plus or minus |
| **×** | Multiplication; used in dimensions or calculations |
| **°C** | Degree Celsius |
| **Å** | Ångström; unit of length |
| **µg** | Microgram |
| **µg/mL** | Micrograms per millilitre |
| **µL** | Microlitre |
| **µm** | Micrometre |
| **cm** | Centimetre |
| **g** | Gram |
| **mL** | Millilitre |
| **mL/min** | Millilitres per minute; mobile-phase flow rate |
| **mg** | Milligram |
| **min** | Minute |
| **mm** | Millimetre |
| **$m^2/g$** | Square metres per gram; specific surface area |

| | |
|---|---|
| **m/z** | Mass-to-charge ratio |
| **nm** | Nanometre; analytical wavelength |
| **pH** | Measure of acidity or alkalinity |
| **psi** | Pounds per square inch; pressure unit |
| **$r^2$** | Coefficient of determination |
| **tR** | Retention time |
| **v/v** | Volume by volume; solvent-mixture ratio |
| **λ** | Wavelength |
| **σ** | Standard deviation (when used in validation equations) |
| **N** | Number of theoretical plates (chromatographic efficiency) |
| **Rs** | Chromatographic resolution or response ratio, depending on context |
| **ACN** | Acetonitrile |
| **API** | Active Pharmaceutical Ingredient |
| **C8** | Octyl-bonded chromatographic stationary phase |
| **C18** | Octadecyl-bonded chromatographic stationary phase |
| **CE–MS** | Capillary Electrophoresis–Mass Spectrometry |
| **ct-DNA** | Calf Thymus Deoxyribonucleic Acid |
| **DNA** | Deoxyribonucleic Acid |
| **EPA** | Environmental Protection Agency |
| **ESR** | Electron Spin Resonance |
| **FDA** | Food and Drug Administration |
| **GC** | Gas Chromatography |

| | |
|---|---|
| **GC–IR** | Gas Chromatography–Infrared Spectroscopy |
| **GC–MS** | Gas Chromatography–Mass Spectrometry |
| **HPLC** | High-Performance Liquid Chromatography |
| **HPLC–MS** | High-Performance Liquid Chromatography–Mass Spectrometry |
| **HPTLC** | High-Performance Thin-Layer Chromatography |
| **ICH** | International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use |
| **IR** | Infrared |
| **IS** | Internal Standard |
| **LC** | Liquid Chromatography |
| **LC–MS/MS** | Liquid Chromatography–Tandem Mass Spectrometry |
| **LC–NMR** | Liquid Chromatography–Nuclear Magnetic Resonance |
| **LOD** | Limit of Detection |
| **LOQ** | Limit of Quantification |
| **MeOH** | Methanol |
| **MS** | Mass Spectrometry |
| **MS/MS** | Tandem Mass Spectrometry |
| **NMR** | Nuclear Magnetic Resonance |
| **PDA** | Photodiode Array |
| **RF** | Response Factor |
| **RP-HPLC** | Reversed-Phase High-Performance Liquid Chromatography |
| **RSD** | Relative Standard Deviation |

| | |
|---|---|
| **SST** | System Suitability Testing |
| **THF** | Tetrahydrofuran |
| **TLC** | Thin-Layer Chromatography |
| **UHPLC** | Ultra-High-Performance Liquid Chromatography |
| **UV** | Ultraviolet |
| **UV–Vis** | Ultraviolet–Visible |
| **USFDA** | United States Food and Drug Administration |

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Ability to work Individually as well as along with the team.

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Participated in international conference on "Innovation And Future Trends In Pharmaceutical And Biomedical Sciences"

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*Participation*

Prof./Dr./Mr./Mrs./ Miss. Ashuka Singh

has actively participated in the two-day International Conference on

***" Innovation and Future Trends in Pharmaceutical and Biomedical Sciences"***

held at Mahadeva Lal Schroff College of Pharmacy, Aurangabad, Bihar on 23rd - 24th March 2026.

Sarita Singh
Chairperson
**Mrs. Sarita Singh**
MLS College of Pharmacy, Aurangabad, Bihar, India.

Deepak
Conference Organizing Secretary
**Dr. Deepak Kumar Gupta**
Principal
MLS College of Pharmacy, Aurangabad, Bihar, India.

Conference Convener
**Dr. Vikas Kumar Singh**
Executive Director
MLS College Of Pharmacy, Aurangabad, Bihar, India.



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GLOBAL PERSPECTIVE ON NEXT GENERATION PHARMACEUTICAL SCIENCE, TECHNOLOGY AND HEALTHCARE

# CERTIFICATE OF PARTICIPATION

This is to certify that Mr./Ms./Dr. **Ashuka Singh** from **Shambhunath Institute of Pharmacy**

has successfully participated as a delegate and delivered an oral presentation on

**Development and Validation of Stability Indicating RP-HPLC and UV spectrophotometry methods for determination of Lenalidomide in Bulk and Tablets**

in the "e-INTERNATIONAL CONFERENCE on GLOBAL PERSPECTIVE ON NEXT GENERATION PHARMACEUTICAL SCIENCE, TECHNOLOGY AND HEALTHCARE" organised by Kamla Nehru Institute of Management and Technology, Sultanpur, held on 16-17 October 2025.

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# Development and Validation of a Reverse Phase High-Performance Liquid Chromatographic Method for Quantitative Estimation of Lenalidomide in Bulk and Pharmaceutical Dosage Form

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**Abstract**

A rapid, highly sensitive high performance liquid chromatographic method has been developed for the determination of Lenalidomide in bulk drug and in capsules. Lenalidomide was eluted from a Inertsil C8 reversed-phase column with a mobile phase consisting of methanol: water (70:30 v/v) pH 9 at a flow rate of 1 mL/min with UV detection at 395 nm. The retention time for Lenalidomide was 6.64 ± 0.05 min. The linear response (r 2 = 0.9991) was observed in the range of 2 - 12 µg/mL with limits of detection (LOD) and quantification (LOQ) being 0.13 and 0.39 µg/ml, respectively. The method shows good recoveries and intra and inter-day relative standard deviations were less than 2.0%. Validation parameters as specificity, accuracy, ruggedness and robustness were also determined. The proposed method provides accurate and precise quality control tool for routine analysis of Lenalidomide in bulk and in capsule dosage form.

**Keywords:** Lenalidomide, RP-HPLC, Method Development, Method Validation, ICH Guidelines